Execise 1

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December 30, 2025

Calculate the DOS for 2D & 3D Dirac fermion using the Hamiltonians below:

$$H_{2D} = \hbar v_F(k_x\sigma_x + k_y\sigma_y)$$
$$H_{3D} = \hbar v_F(k_x\sigma_x + k_y\sigma_y + k_z\sigma_z)$$

Proof. The DOS is defined as:

$$D(E) = \int \frac{d^d k}{(2\pi)^d}\delta(E - E(k))$$

where d is the dimension (2 or 3), and $E(k)$ is the eigenvalue of the Hamiltonian. For 2D case, the eigenvalue is:

$$E = \pm\hbar v_F\sqrt{k_x^2 + k_y^2} = \pm\hbar v_F k$$

Thus,

$$D(E) = \int \frac{d^2 k}{(2\pi)^2}\delta(E - \hbar v_F k) = \int_0^{2\pi} d\theta \int_0^{\infty} \frac{k dk}{(2\pi)^2}\delta(E - \hbar v_F k)$$

Using the property of delta function, we have:

$$D(E) = \frac{1}{(2\pi)^2} 2\pi \frac{E}{\hbar^2 v_F^2} = \frac{E}{2\pi\hbar^2 v_F^2}$$

For 3D case, the eigenvalue is:

$$E = \pm\hbar v_F\sqrt{k_x^2 + k_y^2 + k_z^2} = \pm\hbar v_F k$$

Thus,

$$D(E) = \int \frac{d^3 k}{(2\pi)^3}\delta(E - \hbar v_F k) = \int_0^{\pi} d\theta \int_0^{2\pi} d\phi \int_0^{\infty} \frac{k^2 dk}{(2\pi)^3}\sin\theta\delta(E - \hbar v_F k)$$

Using the property of delta function, we have:

$$D(E) = \frac{1}{(2\pi)^3} 4\pi \frac{E^2}{\hbar^3 v_F^3} = \frac{E^2}{2\pi^2\hbar^3 v_F^3}$$

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